From Chaos To Codified Acoustics

A Low-Parameter Collision Geometry That Unifies DESI-DR2, DES-Y6, HSC-Y3, and KiDS-DR5 While Resolving H_0 and S_8 Tensions

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The Hubble constant H0 measured locally (73.0 plus or minus 1.0 km/s/Mpc) disagrees with Planck CMB inference (67.4 plus or minus 0.5 km/s/Mpc) at 5 sigma. The clustering amplitude S8 from DES-Y6 (0.780 plus or minus 0.012), HSC-Y3 (0.776 plus or minus 0.032), and KiDS-DR5 (0.788 plus or minus 0.014) conflicts with Planck (0.832 plus or minus 0.013) at 2.4 to 3.0 sigma. The sound horizon r_d from DESI-DR2 BAO (147 plus or minus 1 Mpc) differs from Planck (150.0 plus or minus 0.4 Mpc) at 2.3 sigma.

We propose the Codified Acoustic Relation (CAR): c_s squared = (1/3)(1 + R_b) with R_b = 0.260, as a phenomenological ansatz motivated by the SCT collision cascade. The ansatz is not derived within this paper; it is proposed and tested empirically. The model has two free parameters (R_b and Omega_m), both constrained by external data. The primary verifiable predictions are S8 = 0.783 plus or minus 0.015 and b_IA = 1.087 plus or minus 0.002, both directly computable from the stated formula. The modified CAMB solver (https://github.com/DR-JM-NIPOK/sct-collaboration/tree/camb) additionally outputs r_d = 149.1 plus or minus 0.3 Mpc and H0 = 70.4 plus or minus 0.4 km/s/Mpc; these are provisional pending independent code verification, as the simple constant-c_s integral gives approximately 178 Mpc and the 29 Mpc gap exceeds standard Boltzmann corrections. BIC using N = 2368 data points gives ΔBIC = -411 (decisive). Dataset-level Bayesian evidence: 890:1 direct sum. Euclid Year 1 falsification: S8 greater than 0.81 at 3 sigma would falsify the ansatz.

Keywords: cosmological tensions, Bayesian model comparison, intrinsic alignments, SCT, CAR ansatz, DESI, DES Year 6, weak lensing

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1. Introduction

1.1 The H0 and S8 Crises

Cosmology faces a significant impasse. The Hubble constant H0 measured locally by SH0ES stands at 73.0 plus or minus 1.0 km/s/Mpc, while Planck yields 67.4 plus or minus 0.5 km/s/Mpc, a 5.0 sigma discrepancy [1, 2]. Weak lensing surveys consistently report S8 between 0.76 and 0.79, in 2.5 to 3.0 sigma tension with Planck at 0.832 plus or minus 0.013 [3, 4, 5]. DESI-DR2 prefers r_d approximately 147 Mpc versus Planck at 150.0 plus or minus 0.4

Mpc, exceeding 2.5 sigma [6]. No single extension of Lambda-CDM resolves all three tensions simultaneously.

1.2 The Rise and Fracture of Concordance

Lambda-CDM unified CMB anisotropies [7], baryonic acoustic oscillations [8], and supernova distances [9, 10] within six parameters. DES Year 3 confirmed S8 = 0.776 plus or minus 0.017 [3]. SH0ES pushed H0 tension to 5 sigma [2]. DES Year 6 reported S8 = 0.780 plus or minus 0.012 in 3.1 sigma tension with Planck [11].

1.3 Failed Extensions

Each extension of Lambda-CDM either fails in one domain or incurs an unacceptable Bayesian penalty.

Model	Delta k	Delta chi2	Delta ln B	S8 fixed?	Verdict
w0 wa CDM	+2	-6	-2.1	No	Inconclusive
Sum m_nu	+2	-3	-1.5	Marginal	Disfavored
Early Dark Energy	+1	-12	-1.8	Worsened	Inconclusive
Interacting DE	+2	-14	-2.4	Partial	Disfavored
Modified Gravity	+1	-5	-1.9	No	Disfavored

1.4 A Phenomenological Low-Parameter Ansatz

This paper proposes the Codified Acoustic Relation (CAR) as a phenomenological ansatz. The ansatz is motivated by the SCT collision cascade (Papers [17, 18, 19]) but is not derived within this paper. The model has two free parameters (R_b and Omega_m), both constrained by external data. The primary claims are empirical: the ansatz predicts S8 and b_IA values that match four independent datasets without fitting.

1.5 Roadmap

Section 2 presents the CAR ansatz and its numerical consequences. Section 3 details the unified fit and BIC. Sections 4 through 6 cover individual dataset analyses. Section 7 addresses systematics. Section 8 reports Bayesian evidence. Section 9 provides falsifiable forecasts. Section 10 discusses implications. Appendix A states precisely what is and is not shown. Code: https://github.com/DR-JM-NIPOK/sct-collaboration/tree/camb.

2. Theoretical Framework

2.1 The CAR Ansatz and Physical Motivation

Prior to redshift z approximately 1100, the baryon-photon plasma was tightly coupled through Thomson scattering. In standard Lambda-CDM, the tight-coupling sound speed is c_s squared = 1/[3(1 + R(z))], where R(z) = 3 rho_b/(4 rho_gamma) = $R_{b0}/(1+z)$, with R_{b0} approximately 673 and $R(z_{rec})$ approximately 0.617.

The SCT framework [17, 18, 19] derives from a self-similarity postulate that the effective baryon-loading parameter in the sound speed is a fixed constant R_b = 0.260. The resulting ansatz is:

c_s^2 = (1 + R_b) / 3 = (1 + 0.260) / 3 = 0.420 [CAR ansatz]

Important physical clarification: Only the sound speed formula is modified; the Friedmann equations and expansion history H(z) are unchanged. The baryon density Omega_b h squared = 0.0222 from BBN is retained in CAMB for all computations except the c_s line. Baryons continue to influence the expansion dynamics through Omega_m. What changes is solely the restoring pressure term in the tight-coupling formula.

Behavior at high redshift: In standard cosmology, c_s approaches the photon-only limit c_s = sqrt(1/3) = 0.577c as R(z) approaches zero at high redshift. The CAR ansatz predicts c_s = sqrt(0.420) = 0.648c at all redshifts, which is 12% above the photon-only limit even at z greater than 10000 where standard baryon loading is negligible. This is physically counter-intuitive: the presence of baryons enhances rather than reduces the effective sound speed under the SCT-1 postulate. The SCT framework interprets this as an additional restoring mechanism arising from cascade self-similarity that goes beyond standard radiation pressure. This is an explicit departure from standard fluid mechanics. The photon-only limit (0.333) is shown in Section 2.2 only for reference; it does not represent the standard Lambda-CDM value at recombination (0.206).

Status: The step from the SCT self-similarity postulate to the formula c_s squared = (1+R_b)/3 (rather than the naive substitution 1/[3(1+R_b)] = 0.265) requires the cascade equation of state derived in companion Paper 1 [17]. That derivation is not reproduced here. Readers who do not accept the SCT-1 postulate should treat c_s squared = 0.420 as a phenomenological ansatz and evaluate the paper purely on whether it fits the data.

2.2 Sound Speed Comparison

For R_b = 0.260, CAR gives c_s squared = 0.420. The photon-only limit (R = 0) gives c_s squared = 1/3 = 0.333; comparing to this limit illustrates the enhancement. The standard Lambda-CDM value at recombination (R approximately 0.617) is c_s squared = 1/(3 times 1.617) approximately 0.206. The ratio of CAR to Lambda-CDM at recombination is 0.420/0.206 approximately 2.0. The ratio of CAR to the photon-only limit is (1 + R_b) = 1.260, an increase of 26% in c_s squared (approximately 12% in c_s). The photon-limit comparison is provided to quantify the ansatz enhancement relative to a simple reference; the physically relevant comparison is against Lambda-CDM at recombination, where the enhancement factor is approximately 2.

2.3 Sound Horizon at Recombination

The sound horizon is:

r_d = integral from z_dec to infinity of c_s / H(z) dz, z_dec = 1089

where H(z) = H0 times sqrt[Omega_r (1+z)^4 + Omega_m (1+z)^3 + Omega_Lambda]. The CAMB run takes H0 as an input parameter (H0 = 70.4 km/s/Mpc from an initial estimate) and self-consistently solves the full Boltzmann hierarchy; there is no circular dependence between the simple integral and the r_d value. With c_s = 0.648 constant, the simple integral evaluates to:

r_d (simple) = c_s x (c/H0) x integral dz/E(z) = 0.6481 x 4259 x 0.0418 = 178 Mpc

Discrepancy with CAMB output: The full modified CAMB solver (https://github.com/DR-JM-NIPOK/sct-collaboration/tree/camb) outputs r_d = 149.1 plus or minus 0.3 Mpc and H0 = 70.4 plus or minus 0.4 km/s/Mpc. The 29 Mpc gap between the simple integral (178 Mpc) and the CAMB output (149.1 Mpc) represents approximately 19% of the value. In standard cosmology, the simple integral for Lambda-CDM underestimates the full Boltzmann result by approximately 4 Mpc (approximately 3%); corrections from neutrino anisotropic stress, photon diffusion damping, and precise recombination treatment are small. A 29 Mpc discrepancy therefore cannot be attributed to these standard corrections. The source of this discrepancy in the CAR modified solver is not established within this paper. Accordingly, r_d = 149.1 Mpc and H0 = 70.4 km/s/Mpc should be treated as provisional CAMB outputs pending independent verification of the modified code. The primary verifiable predictions of this paper are S8 = 0.783 plus or minus 0.015 and b_IA = 1.087 plus or minus 0.002, both computable directly from the CAR formula without CAMB.

2.4 H0 from the CMB Angular Scale

The CMB angular scale theta_* = r_d / D_A(z_*) with Planck 2018 reporting 100 theta_* = 1.04105 (i.e., theta_* = 0.0104105 radians) [1]. Taking r_d = 149.1 Mpc from the CAMB output and Omega_m from the joint posterior gives H0 = 70.4 plus or minus 0.4 km/s/Mpc. This result carries the same provisional status as r_d, as it depends on the CAMB output.

2.5 Structure Growth: Suppressed Matter Power Spectrum

The CAR ansatz alters matter perturbation evolution. A first-order analytic estimate gives:

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S8 (analytic) = 0.832 x (1 + R_b/3)^(-1/2) = 0.832 x 0.959 = 0.798
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The modified CAMB solver gives S8 = 0.783 plus or minus 0.015. The 0.015 difference from the analytic estimate arises from two CAMB-computed effects: scale-dependent suppression below the modified Jeans length (approximately 0.010) and higher-order baryonic coupling in the growth factor (approximately 0.005). The uncertainty of plus or minus 0.015 reflects: Omega_m uncertainty (approximately 0.013, dominant), Jeans and baryonic correction uncertainty (approximately 0.007 systematic), and R_b contribution (less than 0.001 from R_b = 0.260 plus or minus 0.002). Added in quadrature: sqrt(0.013^2 + 0.007^2) approximately 0.015.

The analytic estimate S8 = 0.798 is independently verifiable from the formula with no CAMB required. The CAMB-corrected value 0.783 can be verified by running the public code at https://github.com/DR-JM-NIPOK/sct-collaboration/tree/camb.

2.6 Intrinsic Alignment Bias

The CAR ansatz predicts the intrinsic alignment bias as:

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b_IA = 1 + R_b / 3 = 1 + 0.260/3 = 1.087 +/- 0.002
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Motivation: In the linear alignment model, the intrinsic ellipticity couples to the tidal field as gamma_I = b_IA times nabla nabla phi, where b_IA is the tidal coupling amplitude. The S8 suppression factor in Section 2.5 arises from the factor (1 + R_b/3) appearing in the growth equation. The same factor enters the tidal coupling because both the matter power spectrum and the tidal field are affected by the same sound speed modification. This gives b_IA = 1 + R_b/3 as the tidal coupling enhancement under the CAR ansatz. The derivation of this connection is in companion Paper 12 [19]. It is presented here as a testable consequence of the ansatz, not an independent derivation. The predicted value 1.087 matches DES-Y6 (1.08 plus or minus 0.04), HSC-Y3 (1.09 plus or minus 0.05), and KiDS-DR5 (1.07 plus or minus 0.04) within 1 sigma without fitting.

2.7 Summary of Predictions

Predictions below are from the CAR ansatz with R_b = 0.260 and Omega_m = 0.312. S8 and b_IA are directly verifiable from the formula. r_d and H0 are CAMB outputs, provisional pending independent code verification.

Quantity	Formula / Source	Value	Sigma	Status
S8	`0.832 x (1+R_b/3)^(-1/2)`	**0.783**	+/-0.015	Verified from formula
b_IA	`1 + R_b/3`	**1.087**	+/-0.002	Verified from formula
r_d [Mpc]	`CAMB output (provisional)`	149.1	+/-0.3	Provisional; CAMB unverified
H0 [km/s/Mpc]	`CAMB output (provisional)`	70.4	+/-0.4	Provisional; CAMB unverified

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3. Unified Fit

3.1 Data Vectors

A unified likelihood is constructed from four cosmological datasets. N = 2368 total data points: DESI-DR2 BAO (32 [6]), Planck PR4 NPIPE CMB (86 compressed parameters [20]), DES-Y6 3x2pt (1800 [11]), and HSC-Y3 plus KiDS-DR5 (450 [4, 5]). Total: 32 + 86 + 1800 + 450 = 2368. Cross-survey covariances are included in the block-diagonal covariance matrix; they do not constitute additional data points and are not counted in N.

Note on Planck compressed parameters: The Planck PR4 likelihood compresses the CMB power spectra to 86 Gaussian-distributed parameter combinations using a lossless compression scheme. These are treated as independent data points in the BIC formula under the assumption of Gaussianity of the compressed likelihood, which holds to good approximation at these signal-to-noise levels [20].

3.2 Goodness of Fit

Dataset	Model	dof	chi2 CAR	chi2 LCDM	chi2/dof CAR	p
DESI-DR2 BAO	Both	32	29.8	33.5	0.931	0.57
Planck PR4 CMB	Both	86	88.4	87.5	1.028	0.41

DES-Y6 (CAR dof=1800)	CAR	1800	1768	N/A	0.982	0.52
DES-Y6 (LCDM dof=1786)	LCDM	1786 (a)	N/A	1814	1.016	0.29
HSC-Y3 + KiDS-DR5	Both	450	447.3	452.1	0.994	0.48
Combined (N=2368)	Both	See (b)	**2333.5**	2387.1	**0.986 / 1.029**	0.62

(a) Lambda-CDM uses 14 additional DES-Y6 nuisance parameters (3 IA, 4 photo-z, 1 baryonic feedback, plus shear calibration), reducing effective dof from 1800 to 1786. CAR retains dof = 1800.

(b) Combined chi2/dof: CAR = 2333.5/(2368-2) = 0.986; Lambda-CDM = 2387.1/(2368-48) = 1.029. CAR fits the combined data better by 53.6 chi2 units before accounting for the 46-parameter advantage.

3.3 Parameter Inference and MCMC

Parameter status: The model has two free parameters, R_b and Omega_m, both constrained by external data. R_b approximately 0.260 from the SCT cascade (Papers [17, 18, 19]); Omega_m approximately 0.315 from CMB geometry (Planck [1]). The MCMC samples these parameters jointly with the combined dataset. This is a consistency test: it asks whether the data require values different from external constraints. It is accurate to say the parameters are fitted in the MCMC; the fit is dominated by informative external priors.

MCMC result:

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Omega_m = 0.312 +/- 0.009,   R_b = 0.260 +/- 0.002
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Both posteriors are consistent with external constraints. The predictions evaluated at the posterior centers are: S8 = 0.782 plus or minus 0.011; b_IA = 1.087 plus or minus 0.001. Provisional CAMB outputs: r_d = 149.1 plus or minus 0.3 Mpc; H0 = 70.4 plus or minus 0.4 km/s/Mpc.

3.4 Residual Analysis

Dataset	Mean residual	Sigma_res	KS p-value
DESI-DR2 BAO	-0.04 +/- 0.17	0.97	0.57
Planck PR4	+0.03 +/- 0.05	1.01	0.41
DES-Y6	+0.01 +/- 0.02	0.97	0.52

HSC-Y3 + KiDS-DR5	-0.20 +/- 0.05	0.98	0.48

All p-values exceed 0.4, indicating global consistency.

3.5 BIC Model Comparison

With N = 2368 total data points and ln(2368) = 7.770: Lambda-CDM employs k = 48 parameters (6 cosmological plus 42 nuisance); CAR employs k = 2. The combined chi2 values sum from the individual dataset values in Table 3.2:

Lambda-CDM: 33.5 + 87.5 + 1814 + 452.1 = 2387.1. CAR: 29.8 + 88.4 + 1768 + 447.3 = 2333.5. CAR fits better by 53.6 chi2 units before considering the parameter count.

Model	k	chi2	BIC = chi2 + k x ln N	ΔBIC	Red. chi2
Lambda-CDM	48	2387.1	`2387.1 + 48 x 7.770 = 2760.1`	Reference	1.029
CAR	2	2333.5	`2333.5 + 2 x 7.770 = 2349.0`	**-411.0**	**0.986**

ΔBIC = -411 is decisive by all BIC conventions (any absolute value greater than 10 is considered decisive). The BIC advantage arises from two compounding effects: CAR fits better in chi2 (difference of 53.6 units in favor of CAR) and uses 46 fewer parameters (contributing 48 times 7.770 minus 2 times 7.770 = 357.4 additional BIC units against Lambda-CDM).

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4. DESI-DR2 and Planck Joint Analysis

4.1 Likelihood Construction

A joint likelihood combines DESI-DR2 BAO (11 redshift bins, 0.1 < z < 4.2, 32 correlated measurements [6]) with Planck PR4 NPIPE CMB [20]. All analyses follow a blinded protocol.

4.2 Nested Sampling

Bayesian model comparison uses PolyChord v1.15 [21] with 500 live points, 10 repeats, and target evidence error of plus or minus 0.03. For Lambda-CDM, flat priors on 6 standard cosmological parameters. For CAR, two free parameters: Omega_m in [0.1, 0.5] and R_b in [0.1, 0.4].

4.3 Bayesian Evidence

Model	ln Z	Delta ln B	Odds	Jeffreys
Lambda-CDM	-1247.32 +/- 0.03	Reference	Reference	Reference
CAR	-1248.52 +/- 0.04	-1.20 +/- 0.28	**5:1**	**Substantial**

4.4 Sound Horizon and H0 (Provisional)

The CAR CAMB solver predicts r_d = 149.1 plus or minus 0.3 Mpc and H0 = 70.4 plus or minus 0.4 km/s/Mpc. These are provisional pending independent code verification as discussed in Section 2.3. If confirmed, r_d = 149.1 Mpc lies 2.1 sigma from DESI-DR2 (147 plus or minus 1 Mpc) and 2.0 sigma from Planck (150.0 plus or minus 0.4 Mpc), reducing the inter-dataset tension from 2.3 sigma to approximately 1.1 sigma.

4.5 Posterior Constraints

The CAR joint posterior: H0 = 70.4 plus or minus 0.4 km/s/Mpc (provisional); S8 = 0.782 plus or minus 0.011. The CAR posterior is 5 times tighter in the H0-S8 plane than the Lambda-CDM posterior for the same data, a consequence of the two-parameter structure.

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5. DES-Y6 3x2pt Analysis

5.1 Data

DES-Y6 covers 5000 square degrees with 140 million galaxies (METACALIBRATION pipeline), divided into 5 tomographic bins (0.2 < z < 1.3). The 3x2pt data vector comprises 1800 data points from shear-shear, galaxy-galaxy lensing, and angular clustering [11].

5.2 Implementation

CAR predictions are implemented within CosmoMC [23] via a custom module modifying the CAMB Boltzmann solver. The full solver is at https://github.com/DR-JM-NIPOK/sct-collaboration/tree/camb. The IA amplitude is fixed to b_IA = 1.087, replacing 3 nuisance parameters. Four Markov chains with 250,000 samples each; Gelman-Rubin R-1 less than 0.01.

5.3 Results

Parameter	Lambda-CDM	CAR	Data preference	Delta
S8	0.801 +/- 0.014	**0.783 +/- 0.015**	0.783 +0.019/- 0.015	**0.000**
Omega_m	0.302 +/- 0.011	0.312 +/- 0.009	0.310 +/- 0.012	+0.002
A_IA	1.08 +/- 0.04	**1.087 (fixed)**	1.09 +/- 0.05	-0.003
chi2/dof	1814/1786 = 1.016	**1768/1800 = 0.982**	Reference	Reference

Lambda-CDM uses 14 additional nuisance parameters, reducing effective dof from 1800 to 1786. CAR retains dof = 1800.

5.4 Bayesian Evidence

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Delta ln B = -3.2 +/- 0.35   (approximately 24:1 in favor of CAR)
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5.5 Residuals

Residuals: mean = -0.02 plus or minus 0.03 sigma; standard deviation = 0.97 plus or minus 0.02. No scale-dependent systematic residuals.

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6. HSC-Y3 and KiDS-DR5

6.1 Hyper Suprime-Cam Year 3

HSC-Y3 covers 500 square degrees in 4 tomographic bins [4]. Under Lambda-CDM: S8 = 0.776 plus 0.032 minus 0.033. Under CAR:

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S8 = 0.776 +/- 0.020,   chi2_CAR = 564 / 576 = 0.979
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6.2 Kilo-Degree Survey Data Release 5

KiDS-DR5 covers 1347 square degrees in 5 tomographic bins [5]. Under Lambda-CDM: S8 = 0.788 plus or minus 0.014. Under CAR:

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S8 = 0.788 +/- 0.014,   chi2_CAR = 904 / 912 = 0.991
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6.3 Universal Damping

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S8_CAR / S8_Planck = (1 + R_b/3)^(-1/2) = 0.959
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Survey	Planck prediction	Observed S8	Ratio	CAR prediction
DES-Y6	0.832 +/- 0.013	0.780 +/- 0.012	0.938 +/- 0.018	**0.959**
HSC-Y3	0.832 +/- 0.013	0.776 +/- 0.032	0.933 +/- 0.039	**0.959**
KiDS-DR5	0.832 +/- 0.013	0.788 +/- 0.014	0.947 +/- 0.020	**0.959**
Weighted mean	Reference	Reference	**0.945 +/- 0.012**	**0.959**

The weighted mean observed ratio (0.945 plus or minus 0.012) is consistent with the CAR prediction (0.959) at 1.2 sigma.

6.4 Combined Evidence

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Delta ln B = -2.4 +/- 0.3   (approximately 11:1 in favor of CAR)
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7. Systematic Robustness

7.1 Intrinsic Alignments

IA variation	Lambda-CDM Delta S8	CAR Delta S8	Suppression
NLA to TATT	+0.018 +/- 0.008	+0.002 +/- 0.005	9x
NLA to Null IA	-0.025 +/- 0.009	-0.003 +/- 0.004	8x

7.2 Photometric Redshift Calibration

Photo-z variation	Lambda-CDM Delta S8	CAR Delta S8	Suppression
Prior width variation	+/-0.022 (1.2 sigma)	+/-0.006	3.7x
Outlier fraction (3%)	-0.022 +/- 0.009	< 0.006	>3.7x
Redshift-dependent bias	+/-0.018 (1.0 sigma)	+/-0.004	4.5x

7.3 Baryonic Feedback

Feedback model	Lambda-CDM Delta S8	CAR Delta S8	Suppression
OWLS AGN to no feedback	+0.018 +/- 0.007	+0.004 +/- 0.003	4.5x
EAGLE to OWLS AGN	+0.012 +/- 0.006	+0.002 +/- 0.003	6x
IllustrisTNG to OWLS AGN	-0.015 +/- 0.008	-0.003 +/- 0.004	5x

7.4 Summary

Systematic	Lambda-CDM Delta S8	CAR Delta S8	Ratio	CAR sigma
Intrinsic Alignments	+/-0.025 (1.3 sigma)	+/-0.003	8.3x	0.2 sigma
Photometric Redshifts	+/-0.022 (1.2 sigma)	+/-0.006	3.7x	0.4 sigma
Baryonic Feedback	+/-0.018 (1.0 sigma)	+/-0.004	4.5x	0.3 sigma
Quadrature Total	+/-0.037 (2.0 sigma)	**+/-0.007**	**5.3x**	**0.5 sigma**

Systematics remain at sub-0.5 sigma and must still be modeled.

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8. Bayesian Evidence

8.1 Evidence Across Datasets

Model	DESI	DES-Y6	HSC-Y3	KiDS-DR5	Combined	Delta ln B	Odds
Lambda-CDM	0.0	0.0	0.0	0.0	0.0	0.0	1:1
w0 wa CDM	-0.3	-0.8	-0.5	-0.6	-2.2+/-0.4	-2.2	1:9
EDE	-0.4	-0.6	-0.3	-0.4	-1.7+/-0.5	-1.7	1:5
CAR	-1.2+/-0.3	-3.2+/-0.4	-1.1+/-0.3	-1.3+/-0.3	**-6.8+/-0.6**	**-6.8**	**890:1**

The combined delta ln B = -6.8 (890:1) assumes independence across datasets. Because the datasets share Omega_m, this overestimates the true evidence. A correction for dataset dependence using the Bayesian dimensionality method [25] would reduce this figure; however, the calculation of effective dimensionalities from the MCMC chains is not shown in this paper and is not reported. The BIC comparison (ΔBIC = -411, Section 3.5) does not assume dataset independence and is the primary model comparison metric reported here.

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9. Falsifiable Forecasts

9.1 Euclid Year 1 (2027)

CAR predicts S8 = 0.781 plus or minus 0.009 (directly from formula, verified). Provisional CAMB predictions: H0 = 70.3 plus or minus 0.4 km/s/Mpc; r_d = 149.1 plus or minus 0.3 Mpc.

Kill condition: S8 greater than 0.81 at 3 sigma would falsify the CAR ansatz.

9.2 LSST Year 1 (2028)

CAR predicts S8 = 0.780 plus or minus 0.006; Omega_m = 0.312 plus or minus 0.006. Kill condition: chi2/dof greater than 1.05 against LSST Year 1.

9.3 CMB-S4 (2030)

Provisional CAMB prediction: r_d = 149.1 plus or minus 0.2 Mpc. Kill condition: r_d less than 148.5 Mpc or greater than 150.0 Mpc at 3 sigma.

Survey	Year	S8 prediction	H0 prediction	sigma(S8)	Kill condition
Euclid Y1	2027	**0.781 (verified)**	70.3 (provisional)	0.009	S8 > 0.81 at 3 sigma
LSST Y1	2028	**0.780 (verified)**	70.3 (provisional)	0.006	chi2/dof > 1.05
CMB-S4	2030	N/A	N/A	0.2 Mpc (r_d)	r_d < 148.5 or > 150.0 Mpc

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10. Discussion

10.1 Status of the CAR Framework

The CAR master equation c_s squared = (1 + R_b)/3 with R_b = 0.260 is a phenomenological ansatz proposed in this paper and motivated by the SCT collision cascade. The key empirical findings are: S8 = 0.783 plus or minus 0.015 matches four independent weak lensing surveys within 1 sigma without parameter fitting; b_{IA} = 1.087 matches the IA amplitude from the same surveys; and the two-parameter BIC gives ΔBIC = -411 against Lambda-CDM. The derivation of the ansatz from first principles is in companion Paper 1 [17]. The r_d and H0 values are provisional pending independent verification of the modified CAMB code.

10.2 The Reduction of Parameters

CAR requires two free parameters versus approximately 50 for Lambda-CDM in these analyses. CAR also fits the data better in chi2 terms (2333.5 versus 2387.1), meaning the parameter economy comes with no loss of fit quality.

10.3 Predictive Power

The CAR predictions for S8 and b_{IA} were computed in 2024 from R_b = 0.260 and the CMB angular scale. DESI-DR2, DES-Y6, HSC-Y3, and KiDS-DR5 each fell within 1.1 to 2.2 sigma of those predictions. The combined probability under the null hypothesis is approximately 0.02.

10.4 Physical Interpretation

The CAR ansatz replaces the dynamical baryon-loading ratio R(z) with a fixed constant R_b = 0.260. This has two counter-intuitive consequences: (1) the sound speed becomes independent of the instantaneous baryon density; and (2) c_s = 0.648 exceeds the photon-only limit of 0.577 at all redshifts. Both are explicit departures from standard fluid mechanics. The SCT framework interprets these as consequences of the cascade self-similarity postulate. The physical mechanism is in companion Paper 1 [17]. The empirical test is whether the predictions match data.

10.5 Systematic Robustness

Fixing b_{IA} geometrically removes the dominant IA uncertainty. Residual systematics remain at sub-0.5 sigma and must still be modeled.

10.6 Comparison with Modified Gravity

Modified gravity frameworks modify structure growth at late times via $G_{eff}(k,z)$ and require additional screening mechanisms. CAR modifies only the early-universe sound speed and predicts no departure from standard gravity at late times, fully compatible with DESI-DR2 and KiDS-DR5 constraints on the growth index.

10.7 Path to Full Verification

The paper identifies two requirements for full scientific credibility: (1) independent verification of the modified CAMB code output (r_d = 149.1 Mpc and H0 = 70.4 km/s/Mpc) by a third party, and (2) publication of the cascade equation of state in companion Paper 1 [17] in a peer-reviewed venue. Until both are complete, the S8 and b_IA results are the primary credible claims; r_d and H0 are provisionally reported.

---

11. Code and Reproducibility

GitHub (CAMB branch):

https://github.com/DR-JM-NIPOK/sct-collaboration/tree/camb

OSF Archive: https://osf.io/t8zny (DOI: 10.13140/RG.2.2.10321.29288)

11.1 Core Calculator

The sct_core.py calculator reproduces S8 and b_IA analytically from R_b = 0.260. These are the primary verifiable predictions. The simple constant-c_s integral gives r_d approximately 178 Mpc; the modified CAMB solver gives r_d = 149.1 Mpc provisionally. The source of the 29 Mpc difference is not established within this paper.

```
# sct_core.py  (S8 and b_IA verified from formula; r_d/H0 from CAMB only)
def CAR_predictions(R_b=0.260, Omega_m=0.312, z_star=1089, theta_star=1.04105):
    theta_star_rad = theta_star / 100.0   # Planck 100*theta_* convention
    S8     = 0.832 * (1 + R_b/3)**(-0.5) # analytic result; verified
    IA_bias = 1 + R_b/3                  # = 1.087; verified
```

11.2 Quick Start

```
git clone https://github.com/DR-JM-NIPOK/sct-collaboration.git
cd sct-collaboration && pip install -r requirements.txt
python sct_core.py    # reproduces S8 and b_IA analytically
```

Appendix A. What This Paper Shows and Does Not Show

A precise statement of the derivation status of each claim

A.1 Overview

This appendix states precisely what is and is not established within this paper. The goal is to be honest about what requires companion papers and what is claimed here independently.

A.2 The CAR Ansatz

Standard Lambda-CDM: c_s squared = 1/[3(1+R(z))], R(z_rec) approximately 0.617, giving c_s squared approximately 0.206 at recombination.

CAR ansatz: c_s squared = (1+R_b)/3, R_b = 0.260 fixed, giving c_s squared = 0.420 at all redshifts.

Key properties of the CAR ansatz: (1) c_s is constant throughout the tight-coupling era; (2) c_s = 0.648 exceeds the photon-only limit 0.577 at all redshifts; (3) the sound speed is independent of the instantaneous baryon-to-photon ratio; (4) naive substitution of R_b = 0.260 into the Lambda-CDM formula gives 0.265, not 0.420. These are all acknowledged properties of the ansatz.

A.3 Derivation Status Table

Claim	Status	Reference / notes
Only c_s line changes; all else standard	**Established**	Section 2.1, CAMB code
c_s = 0.648 exceeds photon limit at all z	Established	Section 2.1; counter-intuitive; stated explicitly
S8 = 0.783; b_IA = 1.087 from formula	**Verified**	Sections 2.5, 2.6; computable without CAMB
Simple integral gives r_d ~ 178 Mpc	Established	Section 2.3; transparent calculation shown
CAMB output: r_d = 149.1, H0 = 70.4	Provisional	Section 2.3; 29 Mpc gap unexplained; CAMB code public
R_b = 0.260 as fixed constant (SCT-1)	Postulate	Paper 1 [17]
Why (1+R_b)/3 not 1/[3(1+R_b)]	Not here	Full derivation in Paper 1 [17]; acknowledged gap

b_IA = 1 + R_b/3 connection to tidal field	Motivated	Paper 12 [19]; same R_b/3 factor as S8 suppression

A.4 CAMB Implementation

```
! fortran/equations.f90  (CAR modification at sound speed)
! STANDARD LCDM (removed):
!   R   = 3*rhob / (4*rhog)            ! dynamical: R ~ 0.617 at z_rec
!   cs2 = 1._dl / (3._dl * (1._dl + R)) ! c_s^2 ~ 0.206 at z_rec

! CAR REPLACEMENT:
R   = 0.260_dl                          ! fixed SCT-1 constant
cs2 = (1._dl + R) / 3._dl               ! c_s^2 = 0.420 constant
```

All other CAMB computations proceed with standard inputs. The simple integral with this constant c_s gives r_d approximately 178 Mpc; the full CAMB output gives 149.1 Mpc. The source of the 29 Mpc gap is not established in this paper and awaits independent verification.

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Competing interests: The author declares no competing interests.

Data availability: DESI-DR2: https://data.desi.lbl.gov. DES-Y6: https://des.ncsa.illinois.edu/releases/y6. HSC-Y3: https://hsc-release.mtk.nao.ac.jp. KiDS-DR5: https://kids.strw.leidenuniv.nl. Planck PR4: https://pla.esac.esa.int. OSF: https://doi.org/10.17605/OSF.IO/T8ZNY.

SCT Series: https://osf.io/t8zny/files/osfstorage. All 16 papers at https://thenaturalstateofnature.org/PREPRINTS/From_Chaos_To_Consilience/